Unit: Unit 1: Ratio, Rate, Percentage
Topic: Rates Variation

Name: ________________
Date: ____________

Quiz (Carolina Reaper)

Q1. A cargo ship travels 120 miles in 4 hours. What is its unit rate in miles per hour?
(A) 20 mph
(B) 30 mph
(C) 40 mph
(D) 50 mph
(E) 60 mph

Q2. If a truck driver can deliver 5 packages in 2 hours, how many packages can be delivered in 10 hours at the same rate?
(A) 20 packages
(B) 25 packages
(C) 30 packages
(D) 35 packages
(E) 40 packages

Q3. A traffic flow model shows that the number of cars passing through an intersection varies directly with the time of day. If 100 cars pass through in 2 hours, how many cars will pass through in 5 hours?
(A) 125 cars
(B) 150 cars
(C) 200 cars
(D) 250 cars
(E) 500 cars

Q4. A shipping company charges

$$0.50$$

per pound to deliver packages. What is the total cost to deliver a 10-pound package?
(A)
$$2.50$$
(B)
$$5.00$$
(C)
$$10.00$$
(D)
$$15.00$$
(E)
$$20.00$$

Q5. A bike messenger travels 12 miles in 2 hours. What is the messenger's unit rate in miles per hour?
(A) 4 mph
(B) 6 mph
(C) 8 mph
(D) 10 mph
(E) 12 mph

Q6. If a plane flies 480 miles in 4 hours, how many miles will it fly in 7 hours at the same rate?
(A) 840 miles
(B) 960 miles
(C) 1080 miles
(D) 1200 miles
(E) 1440 miles

Q7. A freight train can carry 20 tons of cargo per car. If the train has 15 cars, what is the total amount of cargo it can carry?
(A) 200 tons
(B) 250 tons
(C) 300 tons
(D) 350 tons
(E) 400 tons

Q8. A bus travels 25 miles in 1 hour. What is its unit rate in miles per hour?
(A) 20 mph
(B) 25 mph
(C) 30 mph
(D) 35 mph
(E) 40 mph

Open Ended Questions (FRQ)

Q9. A delivery company has two types of trucks: one travels at 40 mph and the other at 60 mph. If the company needs to deliver 200 packages, with each truck able to carry 20 packages, and the distance to the delivery location is 120 miles, how many hours will it take to deliver all packages using both trucks?

Q10. A traffic engineer is designing a new highway system. The engineer wants to ensure that the traffic flow rate is constant throughout the day. If the traffic flow model shows that 150 cars pass through the highway in 3 hours during peak hours, what is the unit rate of cars per hour? Additionally, if the highway has a capacity of 500 cars per hour, what percentage of the highway's capacity is being utilized during peak hours?

Chef's Tips

- Check calculations carefully
- Consider using real-world examples to illustrate concepts
- Emphasize the importance of unit rates in transportation and logistics